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THE SINGLE ENTITY PART B

Reading: Chapter 3 (pages 72-80, but not DELETE and UPDATE on page 79)

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Session Objectives

- Learn to use SELECT sub-clauses on a single table

WHERE Clause Revisited

- Formal syntax of the WHERE clause

  WHERE *predicate*

- Predicates
  - Relational operations (e.g., pe < 12)
  - IN / NOT IN
  - BETWEEN
  - LIKE
  - IS NULL
  - IS NOT NULL

WHERE Predicate Rules

- Predicate relational operators: =, <>, >, >=, <, <=
- The keywords AND and OR - used to combine two predicates into a new one.
- For multiple combinations - parentheses can be used to group combinations
- Without parentheses, the AND operator has a higher precedence than OR

IN Predicate

- Find any values existing in a set of candidates
- Used with a list of values

```
SELECT * FROM Shares WHERE code IN
('FC','AR','SLG');
```

or (using predicate relational operators)

```
SELECT * FROM Shares WHERE code = 'FC' OR
  code = 'AR' OR code = 'SLG';
```

Query-IN

ID	code	shareName	price	quantity	dividend	pe
1	FC	Freedonia Copper	$27.50	10,529.00	$1.84	16
3	AR	Abyssinian Ruby	$31.82	22,010.00	$1.32	13
4	SLG	Sri Lankan Gold	$50.37	32,868.00	$2.68	16

NOT IN

- Not in a list of values

```
SELECT * FROM Shares
WHERE code NOT IN ('CS', 'PT');
```

is equivalent to:

```
SELECT * FROM Shares WHERE code <> 'CS'
AND code <> 'PT';
```

ID	code	shareName	price	quantity	dividend	pe
1	FC	Freedonia Copper	$27.50	10,529.00	$1.84	16
3	AR	Abyssinian Ruby	$31.82	22,010.00	$1.32	13
4	SLG	Sri Lankan Gold	$50.37	32,868.00	$2.68	16
5	ILZ	Indian Lead & Zinc	$37.75	6,390.00	$3.00	12
6	BE	Burmese Elephant	$0.07	154,713.00	$0.01	3
7	BS	Bolivian Sheep	$12.75	231,678.00	$1.78	11
8	NG	Nigerian Geese	$35.00	12,323.00	$1.68	10
10	ROF	Royal Ostrich Farms	$33.75	1,234,923.00	$3.00	6

Ordering Output

- Ordering columns
  - Columns are reported in the order specified in the SQL command

SELECT shareName, pe FROM Shares;

- Ordering rows
  - Rows are ordered using the ORDER BY clause
  - Specify the order with:
    - ASC (ascending) – default
    - DSC (descending)

ORDER BY pe DESC, shareName ASC

Ordering Rows Example

- List all firms where PE is at least 12, and order the report in descending PE. Where PE ratios are identical, list firms in alphabetical order.

```
SELECT * FROM Shares
WHERE pe >= 12
ORDER BY pe DESC, shareName;
```

Be sure to use the correct type for each attribute – to ensure correct ordering

ID	code	shareName	price	quantity	dividend	pe
1	FC	Freedonia Copper	$27.50	10,529.00	$1.84	16
4	SLG	Sri Lankan Gold	$50.37	32,868.00	$2.68	16
9	CS	Canadian Sugar	$52.78	4,716.00	$2.50	15
3	AR	Abyssinian Ruby	$31.82	22,010.00	$1.32	13
5	ILZ	Indian Lead & Zinc	$37.75	6,390.00	$3.00	12

Derived Data

Note the naming of the calculated column

```
SELECT shareName, price, quantity,
  (dividend/price)*100  AS yield
FROM Shares;
```

shareName	price	quantity	yield
Freedonia Copper	$27.50	10,529.00	6.69
Patagonian Tea	$55.25	12,635.00	4.52
Abyssinian Ruby	$31.82	22,010.00	4.15
Sri Lankan Gold	$50.37	32,868.00	5.32
Indian Lead & Zinc	$37.75	6,390.00	7.95
Burmese Elephant	$0.07	154,713.00	14.29
Bolivian Sheep	$12.75	231,678.00	13.96
Nigerian Geese	$35.00	12,323.00	4.80
Canadian Sugar	$52.78	4,716.00	4.74
yal Ostrich Farms	$33.75	1,234,923.00	8.89

How would you change the format of yield?

Built-in Functions

- COUNT
- AVG
- SUM
- MIN, and
- MAX

Functions reduce the dimension of the resulting table

```
SELECT AVG(dividend) AS averageDividend,
AVG((dividend/price)*100) AS averageYield
FROM Shares;
```

The number of rows are reduced consistently across all columns

averageDividend	averageYield
$2.03	7.53

GROUP BY

- What if we apply functions to some, but not all columns?
- GROUP BY is used to project rows with a common value into a smaller set of rows
  - Consistent with row reduction of functions
  - Allows functions to be applied to groups of rows
- Dimension of rows should be consistent across columns
- WHERE clause is applied before the GROUP BY clause

GROUP BY Example

- Average budget of awards, listed by sponsor

```
SELECT sponsorName, AVG(budget)
  AS averageBudget
FROM Awards
GROUP BY sponsorName
ORDER BY AVG(budget) DESC;
```

Note the dimension of output table is consistently reduced by GROUP BY and aggregation

sponsorName	ave
National Institute Occupational Safety & Health	$
Dormitory Authority of the State of New York	$
Empire State Development Corp	$
Long Island Power Authority	$2,822,638.00
Westat Corporation	$2,483,644.02
US Department of Energy	$2,176,573.55
US Army Humphreys Engineer Center Support Activity	$1,383,176.00
US Department of Homeland Security	$1,375,882.00
National Oceanic and Atmospheric Administration	$1,343,209.40

DISTINCT

- Optional keyword in SELECT
- Eliminates duplicate rows in the output table
- Not performed unless requested (for performance reasons)

```
SELECT DISTINCT pe FROM Shares;
```

pe
3
6
10
11
12
13
15
16

Notice that 16 only appears once

Are We on Track?

- In your Awards_SBU DB, extend the previous query to display the list of sponsor names (and maximum budget) with at least one award greater than $1,000,000
- There should be no duplicates in the list
- Sponsor names should be in alphabetical order

sponsorName	maxBudget
Biophotonics Corporation	1,283,334
Brookhaven Science Associates LLC	1,542,546
Centers for Disease Control and Prevention	1,579,365
Dormitory Authority of the State of New York	35,000,000
Empire State Development Corp	47,764,000
Eunice Kennedy Shriver National Institute of Child	1,243,444
Gynecologic Oncology Group	1,040,371
Howard Hughes Medical Institute	1,777,956

Approach to SQL Coding

- An SQL SELECT statement transforms your table into an output table step by step
- Try writing your SQL a step at a time, and check if at each step the output table looks correct

SQL Syntax

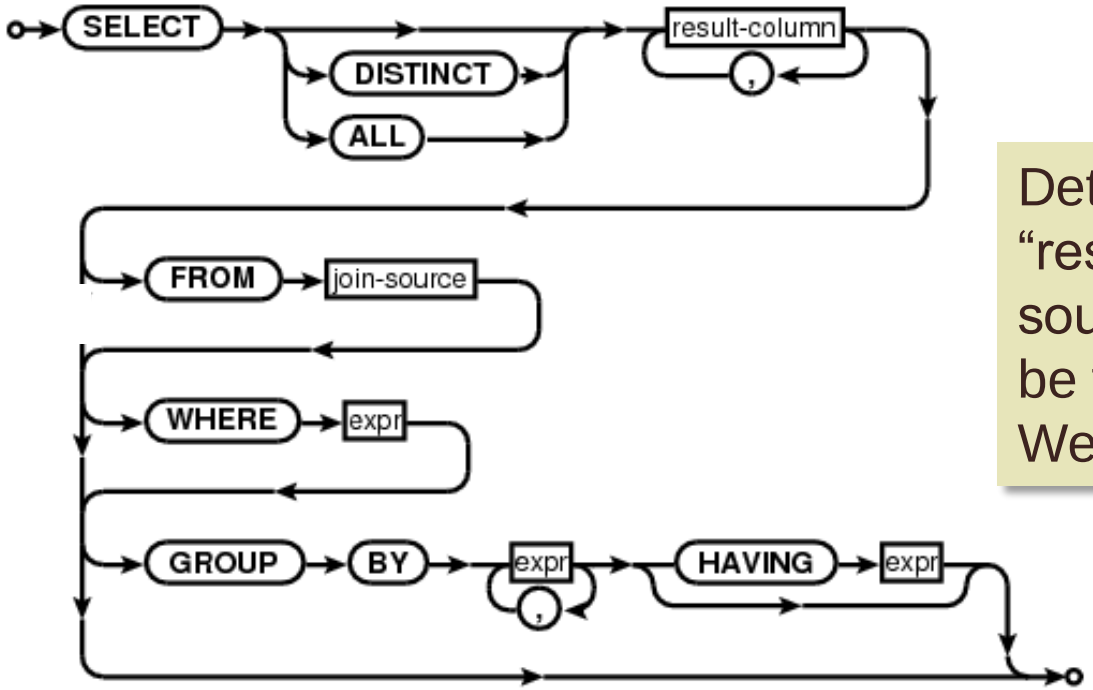
- Now that we've looked at a few SQL statements, let's look more closely at the syntax
  - SELECT is a SQL statement (others include UPDATE, DELETE, DROP TABLE, etc.) to extract info from a DB
  - Semicolon is usually used to separate SQL statements
  - SELECT statement includes optional clauses

sponsorName	averageBudget
National Institute Occupational Safety & Health	$28,055,460.00
Dormitory Authority of the State of New York	$18,271,000.00
Empire State Development Corp	$15,228,624.50
Long Island Power Authority	$2,822,638.00
Westat Corporation	$2,483,644.02
US Department of Energy	$2,176,573.55
US Army Humphreys Engineer Center Support Activity	$1,383,176.00
US Department of Homeland Security	$1,375,882.00
National Oceanic and Atmospheric Administration	$1,343,209.40

```
SELECT sponsorName, AVG(budget)
  AS averageBudget
FROM Awards
GROUP BY sponsorName
ORDER BY AVG(budget) DESC;
```

Flow of Clauses

- Diagram shows the simplified ordering of clauses in a SELECT statement



Detailed syntax for "result-column," "join-source," and "expr" can be found on the SQLite Web page

Source: SQLite

SQL Clauses

SQL Clause	What it Does	Required
SELECT	Lists the fields that contain data of interest.	Yes
FROM	Lists the tables that contain the fields listed in the SELECT clause.	Yes
WHERE	Specifies field criteria that must be met by each record to be included in the results.	No
ORDER BY	Specifies how to sort the results.	No
GROUP BY	In a SQL statement that contains aggregate functions, lists fields that are not summarized in the SELECT clause.	Only if there are such fields
HAVING	In a SQL statement that contains aggregate functions, specifies conditions that apply to fields that are summarized in the SELECT statement.	No

Source: Microsoft

SQL

- The result of every SQL statement is a single output table
- SQL ignores breaks, but it is good to start each sub-clause (e.g., FROM) on a new line
- Keywords are case insensitive, but the convention is to capitalize keywords (e.g., FROM)
- A column used in a WHERE clause is not required to appear in the SELECT column list

HAVING Clause

- Filters rows resulting from a GROUP BY
- Aggregate function used

Example: restrict the display of orgNames to those organizations with more than 60 awards

orgName	totalAwards
050 Interschool Unit Programs	107
050 Chemistry	83
050 Physics	74
050 Pediatrics	64
050 Computer Science	63

```
SELECT orgName, Count(awdNumber)
AS totalAwards
FROM Awards
GROUP BY orgName
  HAVING COUNT(awdNumber)>60
ORDER BY Count(awdNumber) DESC;
```

HAVING Clause

- HAVING is the WHERE clause for groups
- HAVING clause designed for use with GROUP BY to restrict groups that appear in final result table
- Column names in HAVING clause must also appear in the GROUP BY list or be contained within an aggregate function

HAVING applies to the reduced rows, not the original rows

Subqueries

- A query can be used within a query
- Example - report all firms with a PE ratio greater than the average for the portfolio.

WHERE predicate relation is calculated

```
SELECT shareName, pe FROM Shares
WHERE pe > (SELECT AVG(pe) FROM Shares);
```

A query within a query

shareName	pe
Freedonia Copper	16
Abyssinian Ruby	13
Sri Lankan Gold	16
Indian Lead & Zinc	12
Canadian Sugar	15

Notice that the dimensions match

LIKE - Pattern matching

- Example - List all firms containing ‘Ruby’ in their name.

```
SELECT shareName FROM Shares
        WHERE shareName LIKE '*Ruby*';
```

shareName
Abyssinian Ruby

Wildcard character varies in different versions of SQL

LIKE - Pattern matching

- Options
  - Complete field – no wildcard
  - Substrings – wildcard *
  - Single wildcard - ?
  - Sequences
  - Ranges P[A-F]
  - Single digit - #

Are We on Track?

Use SQL View
(not Design View)

- Write a query for the Awards DB to
  - Display organization, sponsor, and number of awards from that sponsor
  - For organizations with 10 or more awards from a sponsor
  - Order the results by organization and award count (largest to smallest)

orgName	sponsorName	numAwards
050 Biochemistry and Cell Biology	National Institute of General Medical Scien	19
050 Chemistry	Brookhaven Science Associates LLC	20
050 Chemistry	National Science Foundation	20
050 Computer Science	National Science Foundation	35
050 Geosciences	NASA Goddard Space Flight Center	11
050 Interschool Unit Programs	National Science Foundation	24
050 Marine Sci Research Center	National Science Foundation	11
050 Materials Science and Engineering	Brookhaven Science Associates LLC	12
050 Mathematics	National Science Foundation	27
050 Molecular Genetics and Microbiology	National Institute of Allergy & Infectious Di	10
050 Office of the Dean of Social Welfare	SUNY at Stony Brook	10
050 Office of the Dean SOMAS	National Science Foundation	18
050 Pediatrics	United Way of Long Island	11
050 Physics	US Department of Energy	17
050 Physics	National Science Foundation	17
050 Physics	Brookhaven Science Associates LLC	14

Did You Meet Session Objectives?

- Learn to use SELECT sub-clauses on a single table